Cognitive Scaffolding in Deep Time: A Research Proposal to Apply Conceptual Integration Theory to Palaeolithic Art

Abstract

This proposal outlines a research program to apply Conceptual Integration Theory (CIT), specifically the model of double-scope blending, to the analysis of Upper Palaeolithic art. The "creative explosion" of this period, marked by the appearance of sophisticated symbolic artifacts, has long been a subject of interpretive debate. Traditional approaches, while valuable, have often focused on discerning the *meaning* of the art, a goal hampered by the vast temporal and cultural gulf separating us from its creators. This research shifts the analytical focus from semiotics to cognition, asking not "What does this art mean?" but "What cognitive operations were necessary for its creation?". We posit that the emergence of therianthropic and other composite imagery, exemplified by the *Löwenmensch* figurine and the "Sorcerer" of Les Trois Frères, provides direct material evidence for the uniquely human capacity for double-scope blending—the integration of conceptually clashing mental frames. By deconstructing these artifacts as products of specific blending operations, this research aims to reconstruct the cognitive architecture of the Palaeolithic mind, frame the Upper Palaeolithic cultural shift as a consequence of this cognitive leap, and establish a rigorous, falsifiable methodology for the field of cognitive archaeology.

Introduction: The Enduring Enigma of the Palaeolithic Mind

Framing the Central Problem

The Upper Palaeolithic period, spanning roughly 50,000 to 12,000 years Before Present (BP), represents a pivotal threshold in human evolution.[1] Archaeologically, this era is distinguished by an unprecedented efflorescence of symbolic behavior, often characterized as a "creative explosion" or a "big bang" of human culture.[2] The material record, particularly in Europe, reveals the appearance of sophisticated blade-based lithic technologies, complex bone and antler tools, widespread use of personal ornamentation, and, most strikingly, the development of complex figurative and abstract art.[4] This suite of innovations signals a fundamental transition in human cognition and is widely associated with the dispersal of anatomically modern *Homo sapiens*.[1]

Despite the richness of this record, its products have resisted conclusive interpretation. The primary challenge is the immense temporal and cultural distance that separates contemporary researchers from the Palaeolithic artists.[7] Without written records or direct ethnographic continuity, any attempt to reconstruct the specific beliefs, myths, or "meanings" behind the art remains profoundly speculative.[8] The core question—why these images were created in deep, often inaccessible, caves—is an enigma that has generated a century of scholarly debate.[8]

Survey of Historical Interpretive Frameworks

The history of Palaeolithic art interpretation reflects an evolving scholarly effort to bridge this interpretive chasm. Early hypotheses, such as "art for art's sake," which posited a purely aesthetic motivation, are now largely dismissed for failing to account for the structured and often hidden placement of the art.[11] More systematic frameworks have since emerged, each providing a distinct lens through which to view this ancient symbolic world.

The first major explanatory model was the theory of **sympathetic magic**, championed by Abbé Henri Breuil. Drawing on early ethnographic analogies, Breuil argued that Palaeolithic art was a form of ritual technology. Depicting an animal was a way to gain magical control over it, thereby ensuring a successful hunt or promoting the fertility of game species.[13] Figures like the "Sorcerer" of Les Trois Frères were interpreted as shamans or magicians presiding over these magical rites.[15]

A significant shift occurred in the mid-20th century with the work of André Leroi-Gourhan, who introduced a **structuralist** approach. Rejecting the idea that images were isolated magical acts, Leroi-Gourhan conducted statistical analyses of the placement and association of figures within caves. He argued that the art was not randomly distributed but formed a

highly organized symbolic system, a "mythogram," structured around a fundamental binary opposition of male and female principles, which he identified with specific animal pairings, most notably horse and bison.[11] Although Leroi-Gourhan himself was hesitant to fully embrace the structuralist label, his work established the crucial premise that the art constituted a coherent, structured system of thought.[17]

The most influential contemporary theory is the **shamanistic hypothesis**, developed by Jean Clottes and David Lewis-Williams. This model integrates neuropsychology, ethnography, and archaeological evidence to propose that much of Palaeolithic art was created within a shamanistic worldview.[19] They argue that the cave itself was conceived as a membrane between the material world and a spirit realm. Shamans, in altered states of consciousness, would enter these deep spaces to make contact with spirits, and the images they painted were manifestations of their visions, often inspired by the natural contours of the rock face.[20]

The Methodological Impasse and the Need for a Cognitive Turn

These frameworks, from the concrete causality of sympathetic magic to the systemic logic of structuralism and the neuro-cognitive basis of shamanism, have each contributed valuable insights. However, they share a common methodological limitation: they primarily attempt to reconstruct a belief system—the *what* of Palaeolithic thought—without first establishing a robust, falsifiable model of the underlying cognitive capacities that made such thought possible—the *how*. This has led to an interpretive impasse, where competing narratives are difficult to test or falsify against the material record alone.[8]

This research proposal argues for a fundamental methodological reorientation. Instead of beginning with the question of meaning, it begins with the question of cognition. By applying a formal model of cognitive operations—Conceptual Integration Theory—this project will analyze Palaeolithic art not as a text to be deciphered, but as a material trace of the cognitive processes that produced it. This "cognitive turn" shifts the focus from reconstructing a lost symbolic content to reconstructing the cognitive architecture of the mind that created it. This approach promises a more objective and scientifically rigorous framework for understanding the profound evolutionary leap that occurred in the Upper Palaeolithic.

Part I: A Cognitive Toolkit for Reconstructing Prehistoric Thought

To reconstruct the cognitive operations of the Palaeolithic mind, it is necessary to establish a theoretical framework grounded in modern cognitive science. This framework is built hierarchically, starting from the fundamental role of the body in shaping thought, moving to the basic mechanisms of conceptual mapping, and culminating in the advanced cognitive operations that enable human creativity.

1.1 From Body to Mind: The Embodied Basis of Abstract Thought

The theory of **embodied cognition** posits that the mind is not a disembodied computer but is fundamentally shaped by the nature of our physical bodies and our sensorimotor interactions with the world.[24] All concepts, even the most abstract, are ultimately physical brain circuits that derive their meaning through neural cascades linked to bodily experience.[27] This perspective is essential for cognitive archaeology, as it provides a basis for understanding the origins of abstract thought in a pre-literate, pre-linguistic, or proto-linguistic context.

The most developed articulation of this principle is **Conceptual Metaphor Theory (CMT)**, pioneered by George Lakoff and Mark Johnson.[29] CMT's central tenet is that metaphor is not merely a device of language but a fundamental mechanism of thought.[28] We understand abstract or less-delineated concepts (the *target domain*) in terms of more concrete, physical, and well-understood concepts (the *source domain*).[32] For example, in the conceptual metaphor ARGUMENT IS WAR, we map concepts from the source domain of WAR (e.g., winning, losing, attacking positions, shooting down arguments) onto the abstract target domain of ARGUMENT.[33]

This process is grounded in **primary metaphors**, which are basic, unconscious neural mappings that arise from recurring correlations in our everyday embodied experience.[27] For instance, the experience of adding more objects to a pile and seeing the level rise creates a physical correlation between quantity and verticality, leading to the primary metaphor MORE IS UP (e.g., "prices *rose*").[27] Similarly, the physical sensation of warmth during affection leads to AFFECTION IS WARMTH, and the experience of difficulty in lifting heavy objects leads to DIFFICULTIES ARE WEIGHTS.[28] Hundreds of these primary metaphors are learned automatically in childhood and form the foundational building blocks of our conceptual system.[27] For Palaeolithic hunter-gatherers, the intense physical and sensory engagement with their environment—tracking animals, throwing spears, navigating landscapes, seeking shelter—would have generated a rich and robust set of primary metaphors, forming the conceptual bedrock upon which more complex symbolic expression could be built.

1.2 The Engine of Meaning Construction: Conceptual Integration Theory (CIT)

While CMT explains how we structure one concept in terms of another, **Conceptual Integration Theory (CIT)**, or "blending," developed by Gilles Fauconnier and Mark Turner, describes the dynamic cognitive operation that allows us to combine concepts to create new meaning.[35] Blending is a ubiquitous and fundamental mental process, on par with analogy and recursion, that operates by selectively combining elements from different mental scenarios into a novel synthesis.[36]

The canonical model of blending involves a network of at least four **mental spaces**, which are small conceptual packets constructed "on the fly" for local purposes of thought and communication.[38]

1. **Input Space 1 & Input Space 2:** These contain the initial concepts or scenarios to be integrated.
2. **Generic Space:** This space captures the abstract structure or frame that the two inputs have in common, which licenses the mapping between them.
3. **Blended Space (The Blend):** This is a new mental space that selectively inherits, or *projects*, some structure from each of the inputs. Crucially, the blend is more than the sum of its parts.

The power of blending lies in its capacity to generate **emergent structure**—new meaning, inferences, and conceptual relationships that are not present in any of the input spaces alone.[36] This emergent structure is created through three key processes within the blend:

- **Composition:** Composing elements from the different inputs to create new relationships. For example, blending a man from one input with a bull from another creates a novel relationship: a man-bull hybrid.
- **Completion:** Bringing in background knowledge and frames associated with the elements in the blend, filling out the scene.
- **Elaboration:** "Running the blend" as a dynamic simulation, exploring its consequences and developing new insights within the blended frame.[38]

A conventional example is the expression, "They dug their own financial grave".[36] This blend takes inputs from the frames of FINANCIAL FAILURE and BURIAL. In the blended space, the emergent structure creates inferences of irreversible, self-inflicted, and final ruin that are far more potent than those available from either input alone.

1.3 The 'Impossible' Idea: Double-Scope Blending as a Marker of Cognitive Modernity

The most complex and powerful form of this cognitive operation is **double-scope blending**. This type of integration is distinguished by its use of inputs whose organizing frames are not only different but are in conflict or clash on vital relations, such as cause-effect, agency, or physical structure.[41] Whereas simpler blends tend to import the organizing logic from one dominant input space, a double-scope blend selectively projects organizing structure from *both* clashing inputs. The result is a highly creative and often paradoxical blended space with a novel organizing frame of its own.[42]

Fauconnier and Turner posit that the capacity for double-scope blending is a uniquely human cognitive singularity that emerged relatively late in our evolutionary history, perhaps between 80,000 and 50,000 years ago.[44] They argue that this cognitive leap was the critical prerequisite for the "human revolution," enabling the suite of behaviors that define cognitive modernity: complex language, art, religion, science, and sophisticated tool use.[44] Double-scope blending is the engine that allows us to conceive of "impossible" things—supernatural beings, abstract scientific concepts, and complex counterfactuals—by forcing together conceptually incongruous domains and creating a new, coherent reality in a blended mental space.[46]

This model provides a specific cognitive mechanism for the broader theory of **"cognitive fluidity"** proposed by archaeologist Steven Mithen.[48] Mithen argues that the modern mind emerged when previously isolated, domain-specific modules of intelligence—such as social intelligence, natural history intelligence (understanding the environment and animals), and technical intelligence (tool-making)—became integrated, allowing for cross-domain thinking.[3] Double-scope blending can be understood as the precise cognitive operation that enables this fluidity. It is the process that allows the frame of SOCIAL intelligence (e.g., personhood, agency, intention) to be blended with the frame of NATURAL HISTORY intelligence (e.g., the form and power of a lion), giving rise to the concept of a supernatural entity with both human-like mind and animal-like power. The appearance of such entities in the archaeological record, therefore, provides a direct material proxy for the presence of this advanced cognitive capacity.

Part II: The Archaeological Evidence for Conceptual Blending

This section applies the cognitive toolkit developed in Part I to the material record of the Upper Palaeolithic. By deconstructing key artifacts as products of conceptual integration networks, we can begin to trace the emergence and development of the cognitive capacities that define the modern human mind. The following table provides a structured overview of the proposed analysis, classifying artworks according to the cognitive operations required for their production.

Table 1: A Comparative Typology of Conceptual Blends in Upper Palaeolithic Art

Artwork /Site	Cultural Period	Est. Date (BP)	Proposed Blend Type	Input Space 1 (Frame)	Input Space 2 (Frame)	Key Projected Elements	Emergent Structure/Meaning
Löwenmensch	Aurignacian	40,000–35,000	Double-Scope	HUMAN	CAVE LION	From 1: Bipedal stance, arms, hands, agency. From 2: Head, powerful shoulders, predatory nature.	A powerful, intentional being transcending the natural order; a master of animals or spirit avatar.
Venus Figurines	Aurignacian/ Gravettian	40,000–25,000	Ontological Metaphor/ Single-	WOMAN	FERTILITY (Abstract)	From 1: Female form. From 2: Exagge	Woman as an entity embodying the

			Scope			ration of fertility-related features (breasts, hips).	abstract concept of fertility; FERTILITY IS A WOMAN.
"Sorcerer"	Magdalenian	~13,000	Double-Scope (Complex)	HUMAN SHAMAN	MULTIPLE ANIMALS (Stag, Bear, etc.)	From 1: Human legs, posture, gaze. From 2: Antlers (stag), paws (bear?), tail (wolf?).	A composite master-spirit embodying the qualities of multiple key animals; a compressed mythogram of the human-animal relationship.
Lascaux Shaft Scene	Magdalenian	~17,000	Double-Scope (Narrative)	HUNTING ACCIDENT	SHAMANIC JOURNEY	From 1: Wounded bison, fallen man. From 2: Bird-he	A narrative blend depicting a physical event (hunt)

						aded man, staff, trance-like state.	as a metaphor for a spiritual event (trance/death/transformation).

This systematic application of the CIT model allows for a structured, comparative analysis of Palaeolithic art across time. It moves the discussion beyond qualitative description toward a formal deconstruction of the cognitive products themselves, making the central thesis of this proposal—that the art reflects the emergence of specific cognitive operations—amenable to rigorous evaluation.

2.1 Case Study I: The Birth of the Supernatural in the Aurignacian

The **Löwenmensch (Lion-man) figurine**, discovered in the Hohlenstein-Stadel cave in Germany, is a seminal artifact for understanding the dawn of cognitive modernity.[49] Radiocarbon dating of the Aurignacian layer in which it was found places its creation between 35,000 and 40,000 years ago, making it the oldest known example of figurative art and the world's oldest known zoomorphic (animal-shaped) sculpture.[49] Carved from the tusk of a woolly mammoth, the 31-centimeter-tall figure required an estimated 400 hours of labor with flint tools, an immense investment of time and skill for a small hunter-gatherer community, signifying its profound cultural importance.[51]

From a cognitive perspective, the *Löwenmensch* is the archetypal example of a **double-scope blend**. It is the material manifestation of a concept that could not exist in the physical world and could only be constructed in the mind through the integration of two radically clashing conceptual frames.

- **Input Space 1: HUMAN.** This input provides the organizing frame of personhood. Key projections from this space include the upright, bipedal stance, the legs and feet, the slender torso, and the arms held at the sides.[51] Most importantly, this input projects the concept of human-like agency, intention, and consciousness.
- **Input Space 2: CAVE LION.** This input provides the organizing frame of the apex predator of the Ice Age European landscape—an animal larger and more powerful than

modern African lions.[51] Projections from this space include the leonine head, the powerful shoulders, and the associated concepts of ferocity, strength, otherness, and dominance in the natural world.

- **The Blended Space:** In the blend, these two incompatible inputs are fused. The clashing body plans (a bipedal human and a quadrupedal lion) are resolved into a single, coherent, and novel entity—a therianthrope.
- **Emergent Structure:** The emergent meaning generated in this blend is that of a being that possesses both human-like consciousness and the supernatural power of the region's most feared predator. This is the cognitive genesis of a supernatural being, an entity that transcends the ordinary categories of the natural world.[51] The creation of the *Löwenmensch* was not merely an act of representation but an act of profound conceptual invention, marking a leap in cognitive ability from perceiving the world to actively constructing new, imaginary worlds in the mind.[52] Furthermore, the choice of material—mammoth ivory—is not incidental. The mammoth was the largest and most powerful herbivore in the environment. Its tusk is not a passive canvas but an active third input into the blend, infusing the final creation with the conceptual weight of another dominant force in the Palaeolithic cosmos. The resulting artifact is thus a three-way blend of [HUMAN + LION] + MAMMOTH, a being of human mind and lion power, embodied in the substance of a mammoth.

2.2 Case Study II: Mythographic Compression in the Magdalenian

Dated to approximately 13,000 BP, the painted and engraved figure known as the **"Sorcerer"** in the Cave of the Trois-Frères in France represents a later, more complex application of the same underlying cognitive operation.[15] Positioned prominently in a chamber dubbed "The Sanctuary," this enigmatic figure has been a focal point of interpretive debate since its discovery.[15] While the accuracy of Henri Breuil's original, widely circulated sketch is contested—particularly regarding the clarity of the antlers and other features—the image is undeniably a therianthropic composite.[15] It appears to combine human legs in a dancing posture and a forward-directed gaze with the features of multiple animals, such as the antlers of a stag, the face of an owl, the paws of a bear, and the tail of a wolf or horse.[57]

The "Sorcerer" can be deconstructed as a complex, multi-input double-scope blend, demonstrating a mature and sophisticated use of this cognitive tool.

- **Inputs:** This is not a simple two-way blend but a "mega-blend" integrating multiple input spaces: HUMAN/SHAMAN, STAG, BEAR, OWL, WOLF, etc.
- **The Blended Space:** The artist has selectively projected key symbolic attributes from each animal onto the central organizing frame of a human figure. The antlers project the concept of the stag's dominance of the forest; the bear's paws project strength; the owl's

face projects vision in darkness; the wolf's tail projects wildness. These are integrated with a human-like posture and an engaging, direct gaze that implies consciousness and communication.
- **Emergent Structure:** The result is not just a hybrid but a composite "master of animals" or a "great spirit".[15] This single figure functions as a form of **mythographic compression**, condensing a complex web of relationships, narratives, and cosmological beliefs about the interconnectedness of the human and animal worlds into one potent, visually coherent image. It embodies an entire cosmology, demonstrating the power of blending to manage and manipulate vast amounts of diffuse meaning into a memorable and powerful conceptual unit.[46]

2.3 A Cognitive Threshold: The Châtelperronian Debate and the Neandertal Mind

The **Châtelperronian** technocomplex, dated to between roughly 45,000 and 40,000 BP and associated with late Neandertal populations, presents a critical test case for this cognitive model.[4] Sites like the Grotte du Renne at Arcy-sur-Cure in France have yielded artifacts, such as perforated animal teeth, ivory beads, and pigment use, that are widely interpreted as evidence for symbolic behavior.[59] This has ignited one of the most significant debates in palaeoanthropology: did Neandertals develop these symbolic behaviors independently, or did they acquire them through acculturation or imitation of newly arrived *Homo sapiens*?.[62]

This debate is complicated by challenges to the stratigraphic integrity of key sites, with some researchers arguing that the symbolic objects are intrusive from overlying Aurignacian layers and thus were not made by Neandertals at all.[59] Others have mounted robust defenses of the association, arguing the symbolic material culture is genuinely Neandertal.[58]

This research proposes to sidestep the "acculturation vs. innovation" debate by asking a more fundamental cognitive question: regardless of who made them and under what influence, do the Châtelperronian symbolic artifacts demonstrate the capacity for double-scope blending? The extant record consists primarily of objects that have been modified or decorated. A perforated fox canine is a symbolic object, likely an ornament. Cognitively, this can be explained as an act of **ontological metaphor** (THIS TOOTH IS A SYMBOL) or a **simplex blend** where an object (tooth) is integrated into a frame (adornment). It does not, however, require the integration of two clashing organizing frames in the manner of the *Löwenmensch*. There are no known Neandertal-associated artifacts that unambiguously represent a therianthropic or similarly "impossible" composite entity.

The stark contrast between the Châtelperronian symbolic toolkit and the sudden appearance

of the *Löwenmensch* in the early Aurignacian suggests a cognitive discontinuity. While Neandertals were clearly capable of symbolic thought, the archaeological record as it currently stands provides no evidence that they possessed the specific and uniquely creative capacity for double-scope blending. This suggests that the consolidation of this cognitive ability in *Homo sapiens* may have been the crucial difference-maker, the cognitive threshold that unlocked the potential for art, religion, and the explosive trajectory of symbolic culture that followed.

Part III: Research Framework and Scholarly Significance

This final section details the proposed methodology for executing this research program, addresses potential challenges and critiques, and outlines the project's broader scholarly significance for the fields of archaeology and cognitive science.

3.1 A New Research Trajectory: From Interpretation to Cognitive Reconstruction

This research is guided by a set of specific questions and testable hypotheses designed to move the study of Palaeolithic art onto a more rigorous cognitive footing.

Research Questions

1. Can the principles of Conceptual Integration Theory, particularly double-scope blending, be systematically applied to create a coherent cognitive typology of Upper Palaeolithic composite art?
2. Does the archaeological record show a diachronic development in the complexity of conceptual blends from the early Aurignacian through the Gravettian, Solutrean, and Magdalenian periods?
3. Does the absence of clear double-scope blends in the Châtelperronian record, when compared to their unambiguous presence in the early Aurignacian, represent a demonstrable cognitive threshold between Neandertals and anatomically modern *Homo*

sapiens?

Hypotheses

- **H1:** Therianthropic and composite figures from the Upper Palaeolithic, such as the *Löwenmensch* and the "Sorcerer," can be formally and consistently deconstructed as double-scope blends with identifiable input spaces, cross-space mappings, and emergent conceptual structures.
- **H2:** Later Magdalenian art (c. 18,000–12,000 BP) will exhibit a higher frequency of more complex blends (e.g., integrating a greater number of input spaces or blending entire narrative frames, as in the Lascaux Shaft Scene) compared to the foundational blends of the earlier Aurignacian (c. 40,000–29,000 BP).
- **H3:** The symbolic artifacts reliably associated with the Châtelperronian can be fully accounted for by simpler cognitive operations (e.g., ontological metaphor, simplex blending) and do not necessitate the invocation of the more complex, frame-clashing operation of double-scope blending.

3.2 Methodological Rigor and Anticipated Challenges

The proposed methodology is designed to be systematic and replicable, while also acknowledging the inherent epistemological challenges of cognitive archaeology.

Methodology

The core of this research will be a systematic survey of the published corpus of Upper Palaeolithic art from Europe, focusing on all known instances of composite or therianthropic imagery. Each artifact or image will be treated as a data point and subjected to a formal cognitive analysis based on the principles of CIT. This analysis will involve:

1. Identifying the constituent parts of the image.
2. Hypothesizing the most plausible input spaces (e.g., HUMAN, LION, STAG).
3. Formally mapping the projected elements from each input to the blended space (the image itself).
4. Identifying the organizing frames of the inputs and determining if they clash (indicating a double-scope blend).

5. Articulating the emergent structure and meaning generated within the blend.

This formal, step-by-step process is designed to constrain subjective interpretation and produce a transparent, replicable analytical procedure that can be evaluated and critiqued by other researchers.

Acknowledging Critiques of Cognitive Archaeology

This project is undertaken with full awareness of the significant critiques leveled against the application of cognitive science to the deep past.

- **The Problem of Subjectivity:** A common criticism of cognitive archaeology is that it can devolve into subjective "just-so stories" where "anything goes".[65] This proposal directly counters this risk by employing a *formal* cognitive theory. Unlike broad interpretive frameworks like "shamanism," CIT has a constrained set of structural rules and principles. An analysis is only valid if it can coherently identify the inputs, mappings, and emergent structure according to the theory's architecture. This provides a level of analytical discipline and falsifiability absent in more purely hermeneutic approaches.
- **The Minimum Necessary Competence Problem:** Archaeology can only ever reveal a fraction of past behaviors, and it is safest to assume that the material record reflects only the *lower boundary* of a past population's cognitive abilities.[67] This research fully embraces this constraint. It does not claim to reconstruct the full spectrum of the Palaeolithic mind or to know what its creators were "thinking." Rather, it seeks to establish the *minimum necessary cognitive competence* required to produce a specific, observable artifact. The production of an object that can be formally deconstructed as a double-scope blend requires, at a minimum, the cognitive capacity for double-scope blending. The art thus serves as a cognitive floor, providing positive evidence for a specific mental operation.
- **The WEIRD Problem and Embodied Cognition:** A valid concern is that many models in cognitive science are based on research with participants from Western, Educated, Industrialized, Rich, and Democratic (WEIRD) societies, who are often cognitive outliers globally.[68] Applying these models to the deep past is fraught with peril. This proposal mitigates this problem in two ways. First, its theoretical foundation rests on embodied cognition and primary metaphors, which are grounded in pan-human sensorimotor experiences that are far more likely to be universal than culturally specific cognitive traits.[24] Second, it explicitly acknowledges the limitations of applying any modern framework to hominins with different bodies and lived experiences, such as Neandertals.[69] The goal is not to project a modern mind onto the past, but to use universal principles of how cognition is scaffolded from embodied experience to reconstruct the evolutionary emergence of a specific, high-level cognitive architecture.

3.3 Conclusion: Re-framing Palaeolithic Art as a Fossil Record of the Modern Mind

This research aims to provide a powerful new explanatory model for a pivotal moment in human evolution. By applying the rigorous framework of Conceptual Integration Theory, it seeks to shift the study of Palaeolithic art from a primarily interpretive discipline, focused on divining meaning, toward a cognitive science grounded in testable hypotheses about mental operations.

The scholarly significance of this project is threefold. First, it offers a novel, systematic methodology for analyzing the cognitive underpinnings of prehistoric symbolic production. Second, it provides a potential resolution to the long-standing debate about the nature of the "creative explosion" by identifying a specific, falsifiable cognitive mechanism—double-scope blending—as a key driver. Third, it reframes the contentious issue of Neandertal symbolism, moving beyond the acculturation/innovation dichotomy to a more fundamental comparison of cognitive capacities as evidenced in the material record.

Ultimately, this research proposes that Palaeolithic art should be viewed as more than just the dawn of aesthetics; it is a fossil record of the modern mind itself. The lines engraved in ivory and the pigments on cave walls are the direct material traces of the novel cognitive operations that allowed our species to imagine new worlds, create supernatural realities, and construct the complex symbolic systems that define us. This project will not claim to reveal the final "meaning" of this ancient art, but it will endeavor to illuminate the very structure of the thought that made meaning possible.

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